26. A posteriori certification of an invariant Hamiltonian torus

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A weakly disturbed mechanical system can sometimes keep moving in an orderly pattern forever, even though the motion never exactly repeats. KAM theory explains how certain such motions survive a perturbation. This problem starts with a proposed orderly motion and checks, with rigorous error bounds, whether a true one must exist nearby. The goal is a guarantee about a specified mathematical orbit family, not a prediction that every nearby trajectory is stable.

Key terms

- **Hamiltonian** — An energy function that generates the system's equations of motion.
- **Integrable system** — A system with enough conserved structure to describe its motion using constant actions and evolving angles in the setting considered here.
- **Action–angle variables** — Coordinates in which the actions label the motion and the angles describe its cyclic phases.
- **Invariant torus** — A torus-shaped set in phase space that trajectories starting on it never leave; it need not be a physical doughnut in ordinary space.
- **Quasiperiodic motion** — Motion combining incommensurate frequencies, so it remains organized without repeating exactly.
- **KAM theory** — Kolmogorov–Arnold–Moser theory, describing persistence of suitable quasiperiodic motions under sufficiently small perturbations.
- **Diophantine condition** — An arithmetic bound keeping integer combinations of the frequencies from becoming too close to zero.
- **A posteriori theorem** — A theorem that uses a computed approximate solution and its error to establish a nearby exact solution.
- **Residual** — The amount by which the approximate solution fails to satisfy the exact equation.
- **Interval arithmetic** — Computation with enclosing intervals, using controlled rounding to bound numerical errors.

Why this matters

Long-term stability is difficult to infer from a finite trajectory simulation. A theorem can instead prove that a particular organized motion exists despite perturbations. This offers a foundation for studying stability mechanisms in celestial and other Hamiltonian systems, provided the model and theorem assumptions are appropriate.

What progress would mean

A certificate for a nonzero perturbation would convert an approximate torus into an existence result. Applying the method to planetary observations would additionally require reduction of the planetary model and a careful treatment of measured initial-condition uncertainty.

Core question

For an explicitly specified analytic near-integrable Hamiltonian and an approximate torus, can every hypothesis of a quantitative KAM theorem be verified?

Scope and assumptions

Begin with $H(I,\theta)=\omega\cdot I+|I|^2/2+\epsilon[\cos\theta_1+\cos(\theta_1-\theta_2)]$ in two degrees of freedom, $\omega=(1,(1+\sqrt{5})/2)$, on a stated complex neighborhood of a real action box. Choose a rational ϵ only after validating the unperturbed case. Seek an embedding with rotation vector ω.

This is a mathematical model, not a solar-system stability assertion. Planetary extensions require an explicit canonical reduction, handling Kepler degeneracy, collision exclusions, and input uncertainty.

Mathematical target

For an approximate embedding $K:T^2\to$phase space define $E(\theta)=X_H(K(\theta))-DK(\theta)\omega$. Select a published a posteriori KAM theorem and reproduce its hypotheses and computable constants: analytic widths, nondegeneracy/twist inverse, residual norm, and small-divisor bounds.

Prove $|k\cdot\omega|\geq\gamma/|k|_1\hat{}\tau$ for every nonzero integer k using arithmetic properties of the exact chosen ω. Finite checks alone are insufficient. Evaluate the theorem's smallness inequalities with outward-rounded intervals and certify closeness of a true K_* to K.

Required outputs and proof obligations

Export H, ϵ, domain, exact frequency representation, Fourier coefficients, all theorem constants, interval inequalities, and a checker. State the conclusion: existence of a nearby quasiperiodic invariant torus with a quantified embedding error. A failed sufficient condition yields "not certified," not "unstable."

Validation and rejection controls

Verify $E=0$ for $K_0(\theta)=(I=0,\theta)$ at $\epsilon=0$. Derive an all-integer Diophantine bound for the golden-ratio frequency. Check Fourier tails and loss of analytic width. Test a resonant frequency where the stated theorem must refuse certification.

Milestones and research extension

Baseline: unperturbed certificate and complete arithmetic/theorem setup. Strong: a nonzero-ϵ torus certificate. Research extension: a larger perturbation range or a reduced planetary model. One invariant torus does not certify a neighborhood of arbitrary observed initial conditions or a phase-space measure bound.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Valvo and Locatelli, Hamiltonian Control of Magnetic Field Lines: Computer Assisted Results Proving the Existence of KAM Barriers — Example of a complete computer-assisted Hamiltonian KAM workflow; use a theorem matching the chosen formulation.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$H(I,\theta) = \omega \cdot I + \frac{|I|^2}{2} + \epsilon[\cos\theta_1 + \cos(\theta_1 - \theta_2)], \qquad \omega = (1, (1+\sqrt{5})/2),$$

$$E(\theta) = X_H(K(\theta)) - DK(\theta)\omega, \qquad |k \cdot \omega| \geq \frac{\gamma}{|k|_1^\tau} \quad (k \in \mathbb{Z}^2 \setminus \{0\}).$$

Prove the all-integer Diophantine bound and every analytic-width, twist, tail and smallness condition of a specified quantitative theorem. Its conclusion is a nearby invariant torus, not stability of arbitrary planetary initial conditions. Failure of a sufficient criterion is not instability.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — finite Diophantine "certificate" (126).** Checking |k|≤K_max cannot prove an infinite small-divisor condition. Rational decimal frequencies are not exact irrational inputs, and a full uncertainty box generally contains resonances.

2. **Major — missing nondegeneracy and reduction (21–25, 74).** The Kepler action formula and derivative are inconsistent, and planetary Kepler Hamiltonians are properly degenerate. A generic nondegenerate theorem cannot be applied without reduction/averaging work.
3. **Major — torus versus actual trajectory (29–40, 113–120).** Surviving tori are deformed embeddings; existence does not place the measured solar system on one. Measure and time claims require additional arguments.
4. **Major — iteration as proof (713–718).** A shrinking residual over a few steps does not establish convergence. The new task ties every computational quantity to one quantitative theorem and allows honest failure of its sufficient conditions.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For an explicitly specified analytic near-integrable Hamiltonian and an approximate torus, can every hypothesis of a quantitative KAM theorem be verified?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: unperturbed certificate and complete arithmetic/theorem setup. Strong: a nonzero-ϵ torus certificate. Research extension: a larger perturbation range or a reduced planetary model. One invariant torus does not certify a neighborhood of arbitrary observed initial conditions or a phase-space measure bound.

Sources supporting the corrections

- Valvo and Locatelli, Hamiltonian Control of Magnetic Field Lines: Computer Assisted Results Proving the Existence of KAM Barriers — Example of a complete computer-assisted Hamiltonian KAM workflow; use a theorem matching the chosen formulation.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search,

and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and

resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.